

Curated Research Articles

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- **Redesigning Interlayers for Anode-Free Batteries: Harnessing Continuity and Metal Inventory** — score: 0.900 The article discusses the strategic redesign of interlayers in anode-free batteries to ensure effective interfacial coverage while optimizing the use of active-metal inventory from the cathode. By integrating thin-film closure models with stoichiometric consumption of metal, the authors identify key design parameters—including morphology, chemistry, and deposition techniques—that are critical for achieving efficient and practical battery performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Delocalized Redox Framework of Indanthrone Enables Low-Strain and Durable $\text{Mn}^{2+}/\text{H}^{+}$ Storage in Aqueous Batteries** — score: 0.900 The study presents an indanthrone cathode that enhances the performance of manganese-based hybrid aqueous batteries by enabling a low-strain and durable redox mechanism. This cathode architecture, characterized by extensive electronic delocalization, allows for a high specific capacity and excellent capacity retention over numerous cycles, advancing the development of safe and cost-effective battery alternatives to lithium-ion systems. Journal: *Wiley: Small Methods: Table of Contents*
- **In-Situ Formed Bionic Cell Wall Interphase Enables Reversible Zn Anode for High-Performance Aqueous Batteries** — score: 0.800 The article discusses the development of a bionic cell wall interphase that enhances the stability and performance of zinc anodes in aqueous zinc-ion batteries (AZIBs). This innovation allows for the formation of a reversible solid electrolyte interphase, thereby addressing common issues faced in battery performance and promoting safer, cost-effective energy storage solutions. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Synergistic Anion Confinement in a Poly(Ionic Liquid)/MOF Composite Electrolyte Decouples Ionic Conductivity and Mechanical Strength for High-Performance Solid-State Lithium Metal Batteries** — score: 0.800 The study introduces a composite poly(ionic liquid) electrolyte that combines a main-chain PIL and the metal-organic framework MOF-5, effectively addressing the trade-off between ionic conductivity and mechanical strength. This novel electrolyte achieves a conductivity of 1.4 mS cm^{-1} , high Li^{+} transference, and excellent electrochemical stability, enabling over 2000 hours of symmetric cell stability and 700 cycles in full cells with LiFePO_4 , demonstrating its potential for high-performance solid-state lithium metal batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **Matching d-p Energy Levels Between High-Entropy Alloy Catalysts and Chalcogen Cathodes for All-Solid-State Batteries** — score: 0.800 The article presents a strategy for enhancing the performance of all-solid-state lithium-chalcogen batteries by matching the d band of high-entropy alloy (HEA) catalysts with the p band of chalcogen cathodes. This approach improves electronic coupling and charge transfer, leading to accelerated solid-state conversion reactions. The study identifies FeCoNiCuCr as a particularly effective catalyst, resulting in a stable reversible capacity of 657 mAh g^{-1} after 1800 cycles, thus addressing kinetic challenges in these energy storage systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Fluorinated Diluents Enable Crowded Solvation Environments to Form Anion-Rich SEIs for High-Performance Potassium-Ion Batteries** — score: 0.800 This study introduces a localized high-concentration electrolyte (LHCE) for potassium-ion batteries, incorporating 1,2-difluorobenzene as a diluent to enhance electrolyte performance. The modified electrolyte fosters anion-rich solid electrolyte interfaces (SEIs) that significantly improve electrochemical kinetics and cycling stability, leading to high capacity and efficiency in potassium-ion battery systems. Journal: *Wiley: Small Methods: Table of Contents*
- **From Material to System: A Review of Self-Charging Aqueous Zinc-Ion Batteries** — score: 0.700 This review explores self-charging aqueous zinc-ion batteries (SC-AZIBs) that combine energy harvesting with storage, enabling usage of ambient energy sources. It analyzes various self-charging

mechanisms, emphasizes design strategies for cathode materials and devices, identifies significant challenges faced by the technology, and proposes future development pathways for enhanced performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Suspension Electrolyte for Protecting Aqueous Zinc Batteries From Calendar Aging** — score: 0.700 This research introduces a novel Al₂O₃ suspension electrolyte designed to enhance the calendar life of aqueous zinc batteries (AZBs) by effectively reducing zinc corrosion during extended idle periods. By integrating a small percentage of Al₂O₃ nanoparticles into a traditional ZnSO₄ electrolyte, the study demonstrates substantial improvements in battery performance, including a threefold increase in lifespan, higher Coulombic efficiency, and sustained specific capacity, making this approach promising for safe and cost-effective large-scale energy storage solutions. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A mechanistic investigation of the relationship between battery energy and capacity** — score: 0.600 The article explores the discrepancy between battery capacity (Ah) and energy (Wh) as indicators of state of health, emphasizing the importance of energy for practical application. It develops a theoretical framework that connects this divergence to various degradation mechanisms influenced by electrode chemistry, operating conditions, and discharge depth, ultimately aiming to enhance lifetime predictions and safety assessments in battery usage. Journal: *Joule*
- **The interplay between stack pressure, mechanical expansion and degradation pathways in lithium-ion batteries** — score: 0.600 This study investigates the effect of stack pressure on the long-term cycling performance of lithium-ion batteries, an area that has not been extensively explored. Utilizing a high-precision pressure-controlled dilatometry tool, researchers aim to optimize stack pressure to enhance battery longevity and better understand its influence on mechanical expansion and degradation pathways. Journal: *Nature Energy*
- **Reconciling triple-phase boundaries and tortuosity in high-energy all-solid-state lithium-sulfur batteries** — score: 0.600 The article presents a design principle for all-solid-state lithium-sulfur batteries that optimally combines the influence of tiny pores on sulfur reactivity with larger carbon particles that facilitate ion transport. This integration addresses the challenges of achieving high sulfur utilization and cycling durability, positioning the approach as a significant advancement for next-generation battery technology. Journal: *Joule*
- **Suppressing the interface electric field fluctuation through faceto-face electronic channel regulation for sustainable zinc batteries** — score: 0.400 The article discusses advancements in sustainable zinc batteries by addressing the challenges posed by poor electrode-electrolyte interface interactions. Researchers propose a method to suppress electric field fluctuations through a face-to-face electronic channel regulation, aiming to improve the effectiveness of gel electrolytes in aqueous zinc systems. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Algae-inspired hydrogel electrolytes with sulfate-group-mediated polyiodide repulsion for stable iodine chemistry** — score: 0.400 Researchers have developed algae-inspired hydrogel electrolytes that utilize sulfate groups to mitigate issues such as zinc dendrite formation and polyiodide shuttling in aqueous zinc-iodine batteries. This innovation aims to enhance the stability and efficiency of iodine chemistry, thereby improving the viability of these batteries for large-scale energy storage applications. Journal: *RSC - Energy Environ. Sci. latest articles*
- **A rechargeable non-aqueous Mg–O₂ battery based on magnesium peroxide chemistry** — score: 0.400 The study demonstrates that by modifying the solvation environment of Mg²⁺ ions, researchers have achieved reversible magnesium peroxide chemistry, leading to stable cycling in a rechargeable non-aqueous magnesium–oxygen battery. This advancement addresses the previously encountered issue of poor reversibility, thereby enhancing the potential for high energy density applications. Journal: *Nature Chemistry*
- **Dendrite initiation and deflection in biaxially stressed solid electrolytes** — score: 0.300 The study presents findings that in-plane biaxial compression influences dendrite propagation in garnet solid electrolytes, revealing that dendrite initiation takes place within the material’s interior over extended

cycling periods. This observation enhances the understanding of dendrite formation mechanisms in solid electrolytes, which has implications for battery performance and longevity. Journal: *Nature*

- **Continuous Fabrication of Dendrite-Free Zinc Fiber Anodes for Flexible Batteries** — score: 0.300 This study presents the development of a scalable fiber-shaped zinc electrode for zinc-ion batteries that minimizes dendrite formation through optimized nucleation and growth techniques using oxygen-functionalized carbon nanotube fibers. The continuous fabrication process yields meter-scale, durable batteries with enhanced corrosion resistance and a notable energy density, positioning them as promising candidates for safe, flexible power sources in wearable electronics. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Critical surface strain regime for stable and active epitaxial platinum oxygen reduction electrocatalysts** — score: 0.200 The study reveals a critical thickness-dependent strain in epitaxial platinum/iridium films that balances improved electrocatalytic activity for the oxygen reduction reaction with long-term stability, addressing the common issue of irreversible strain relaxation in catalyst performance. Journal: *Nature Energy*
- **MXene-based bioabsorbable bandage speeds up diabetic wound repair** — score: 0.000 The article discusses a novel glucose-powered electronic bandage made from MXene-based electrodes that is fully bioabsorbable. This bandage is designed to harness biochemical energy to enhance the healing process of diabetic wounds by restoring the electrical fields at wound edges, thereby accelerating recovery. Journal: *Nature Reviews Materials*
- **Chlorinated Polypropylene Enables Stress-Dissipative Networks and High Efficiency in Intrinsically Stretchable Organic Photovoltaics** — score: 0.000 The article discusses the use of chlorinated polypropylene in enhancing the performance of intrinsically stretchable organic photovoltaics (IS-OPVs) by enabling stress-dissipative networks. This innovation seeks to achieve a balance between high power conversion efficiency and mechanical flexibility, addressing a critical challenge in current IS-OPV materials. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Energy efficient integrated H₂O₂ electrosynthesis via tailored bubble dynamics** — score: 0.000 The article discusses a novel approach to H₂O₂ electrosynthesis that enhances energy efficiency by integrating it with anodic upgrading, enabling a potential for electricity-free operation. The authors focus on overcoming interfacial kinetic bottlenecks at the anode, highlighting key factors that impede progress towards improved performance in this innovative method. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Cost-effective valorization of 2,3-butanediol to high-value chemicals and jet fuel** — score: 0.000 The article describes an optimized process for converting biomass-derived 2,3-butanediol into sustainable aviation fuel, addressing the separation challenges of 2,3-butanediol through a reactive extraction method that uses in situ generated isobutyraldehyde. The approach promises economic feasibility with production costs under \$2.50 per gallon and offers significant environmental benefits, achieving over a 58% reduction in CO₂ emissions. Journal: *Joule*
- **Smart Antibacterial Janus Fabric Based on PVDF/Ag-Decorated-MXene for Unidirectional Water Transport and Thermal Management** — score: 0.000 The study presents PAM-Textile, a multifunctional Janus fabric made from hydrophobic polyvinylidene fluoride (PVDF) and hydrophilic silver-decorated MXene (Ag@MXene), designed for enhanced comfort in wearable textiles. It exhibits efficient unidirectional moisture management and significant thermal regulation, reducing skin temperature by up to 12.1°C, while also demonstrating broad-spectrum antibacterial properties against MRSA. The fabric maintains its functionalities after multiple washes and abrasions, showcasing potential for advanced applications in healthcare and sportswear. Journal: *Wiley: Small Methods: Table of Contents*
- **Fractionation-Free Protein Corona Quantification Through Synchrotron-Based Small-Angle X-ray Scattering** — score: 0.000 This study presents a novel synchrotron-based small-angle X-ray scattering (SAXS) technique that quantifies protein corona formation on silica nanoparticles without disrupting their native environment. By distinguishing between free and bound

proteins, it allows for direct measurement of adsorbed mass in both simple and complex media, and establishes thermodynamic parameters related to protein adsorption behavior. This method enhances the non-invasive characterization of protein coronas, paving the way for better understanding of nanoparticle interactions in biological systems. Journal: *Wiley: Small Methods: Table of Contents*

- **Accelerated Formation of CsPbI₃ through Controlled Amidation of Intermediate Phases** — score: 0.000 This research introduces a novel crystallization technique for creating phase-pure CsPbI₃ thin films by employing 2,2,2-trifluoroethyl trifluoroacetate (TFTE), which swiftly reacts with dimethylammonium iodide (DMAI) to eliminate residues that can degrade film quality. As a result, the resulting solar cells exhibit a remarkable efficiency of 20.8% and enhanced stability, addressing previous challenges in film formation and paving the way for better-performing perovskite photovoltaics. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Giant and Broad-Temperature Electrocaloric Cooling via B-Site Medium-Entropy Design in PST-Based Multilayer Ceramics** — score: 0.000 The study introduces a novel B-site medium-entropy Pb(Sc_{0.25}Ta_{0.25}In_{0.25}Nb_{0.25})O₃ multilayer ceramic capacitor, which significantly enhances electrocaloric cooling by achieving a ΔT_{ad} of 9.34 K over a stable 90 K temperature range with minimal electric field requirements. This advancement is attributed to induced lattice distortions and glassy polar nanoregions, providing a framework for developing efficient solid-state cooling technologies that resist thermal instability and maintain durability through extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Togetherness: How co-operation built the world** — score: 0.000 The article features science journalist Rowan Hooper discussing the concept of symbiosis, emphasizing how collaboration among organisms has played a crucial role in shaping the natural world. It explores the significance of cooperation in ecological and evolutionary contexts, highlighting examples of interdependent relationships in nature. Journal: *Nature*
- **Backreaction of stimulated Hawking radiation in an optical analogue** — score: 0.000 The article explores the generation of Hawking radiation through a fiber-optical system mimicking black hole physics, providing both theoretical insights and experimental validation. It emphasizes the impact of this radiation on the optical pump, thereby illustrating the interplay between stimulated radiation and its backreaction in a controlled setting. Journal: *Nature*
- **Harmonizing standards and resources for the medical genome** — score: 0.000 The article highlights the difficulties and potential in genomic sequencing for various clinical applications, particularly in genetic therapies. It advocates for the creation of centralized reference standards to enhance consistency and accuracy in precision medicine. Journal: *Nature*
- **Towards the construction of a virtual yeast** — score: 0.000 The article discusses the development of a virtual yeast, an AI-based model designed to simulate and analyze the behaviors of eukaryotic cells. This innovative approach aims to enhance understanding of cellular processes and facilitate advancements in biological research and applications. Journal: *Nature*
- **Isomeric multi-hydrogen-bonding enables blue perovskite LEDs** — score: 0.000 The article discusses how the use of isomeric molecules to create hydrogen-bonding networks has led to the development of stable blue perovskite LEDs, achieving unprecedented external quantum efficiencies of 22.0%. This advancement represents a significant step forward in the performance of full-color display technologies. Journal: *Nature*
- **A secreted endosymbiont protein essential for colonizing host cells** — score: 0.000 The study identifies SyeA, a secreted protein from the endosymbiont *Buchnera*, which plays a critical role in the protein's ability to colonize host cells. This protein, present in the ancestor of *Buchnera*, is homologous to effector proteins associated with pathogenic bacteria and is necessary for the successful transmission of *Buchnera* within its host. Journal: *Nature*
- **Replication-stress-induced chromatin loops protect fork stability** — score: 0.000 The article explores how replication stress prompts the formation of temporary chromatin loops that protect stalled

replication forks, which are enriched with new heterochromatin. This mechanism appears to play a crucial role in maintaining the stability of these forks during periods of cellular stress. Journal: *Nature*

- **Identification of cross-stage, cross-species malaria CD8+ T cell antigens** — score: 0.000 The study utilizes immunopeptidomics to uncover a range of conserved Plasmodium T cell antigens that are expressed during various stages of the malaria parasite’s life cycle across different species. These findings highlight potential avenues for malaria vaccine development, emphasizing the importance of targeting these antigens. Journal: *Nature*
- **Targeted enzyme discovery using metal-coordination mining** — score: 0.000 The article presents a novel methodology for exploring protein structure databases, focusing on the unique characteristics of metal-binding active sites in enzymes. This approach has successfully led to the identification of previously unknown families of radical halogenases, expanding the understanding of enzyme diversity and function. Journal: *Nature*
- **Directly probing the carrier transfer length in 2D-material transistors** — score: 0.000 The article investigates the carrier transfer length in bismuth-contacted monolayer MoS2 transistors using cross-sectional scanning tunneling microscopy, revealing a transfer length of just 2 nm. This finding establishes the limitations for metal-contact scaling in future sub-10 nm 2D electronic devices, highlighting challenges in enhancing performance at such small scales. Journal: *Nature*
- **Restoring cortical disinhibition improves Huntington’s disease phenotypes** — score: 0.000 Researchers demonstrate that enhancing cortical disinhibition in a mouse model of Huntington’s disease can effectively restore neuronal network activity and alleviate associated movement impairments, highlighting a potential therapeutic direction for the disorder. Journal: *Nature*
- **Tin perovskite transistors stabilized through volatile coordination** — score: 0.000 The article discusses a novel volatile-assisted coordination method that enhances the stability of tin perovskite semiconductors by effectively managing surface defects and self-doping. This breakthrough allows for the development of transistors that can operate reliably for a month at elevated temperatures of 100°C. Journal: *Nature*
- **Hadean bridgmanite in the source of a present-day ocean island** — score: 0.000 The study presents high-precision neodymium isotope analyses of lavas from the Fani Maoré volcano in the Comoros, revealing the presence of Hadean bridgmanite and highlighting geological heterogeneities that originated in Earth’s formative years. This finding provides insights into the ancient processes that have shaped the Earth’s mantle and current volcanic activity. Journal: *Nature*
- **Steatosis shapes prognosis-defining liver metastasis heterogeneity in CRC** — score: 0.000 The study reveals that liver steatosis enhances the development of replacement liver metastases in colorectal cancer (CRC) by stabilizing the MYC protein via acetylation. This process subsequently boosts proline synthesis and collagen production, suggesting that steatosis plays a critical role in determining the heterogeneity and prognosis of liver metastases in CRC patients. Journal: *Nature*
- **Connecting single-cell transcriptomes to projectomes in the mouse visual cortex** — score: 0.000 The study investigates the relationship between single-cell transcriptomes and the projection patterns of excitatory neurons in the mouse visual cortex. By integrating cell-type definitions, gene expression, and spatial localization, the research reveals how these factors together predict the long-range targets of neuronal connections, thereby linking molecular characteristics to their functional properties within the neural circuit. Journal: *Nature*
- **Dual tumour–myeloid targeting of glioblastoma with GPNMB CAR-T cells** — score: 0.000 The study identifies GPNMB as a common antigen in both glioblastoma tumors and their microenvironment through comprehensive multi-omic profiling. Furthermore, GPNMB-targeted CAR-T cells exhibit significant therapeutic effects in vitro and in animal models, suggesting a promising dual targeting approach for glioblastoma treatment. Journal: *Nature*
- **Heat-triggered phospholipid flipping stabilizes plasma membrane fluidity** — score: 0.000 The study reveals that in rice, the P4-ATPase complex OsALA5–OsALIS2 plays a crucial role in maintaining

plasma membrane fluidity under high temperature by rapidly flipping saturated phosphatidylcholines to the cytosolic side, thereby enhancing the membrane’s adaptability to thermal stress. Journal: *Nature*

- **TROP2 targeting reveals therapy-driven cell state dynamics in colorectal cancer** — score: 0.000 The study investigates the role of TROP2 in colorectal cancer, highlighting its presence in aggressive tumor cells that exhibit resistance to therapy. By targeting TROP2 with antibody-drug conjugates alongside chemotherapy, researchers found improved treatment outcomes, suggesting a modulation of cellular plasticity that helps combat resistance in cancer treatment. Journal: *Nature*
- **Aerosols and hydrocarbons in the atmosphere of a white dwarf planet** — score: 0.000 The study uses transmission spectroscopy from the James Webb Space Telescope to analyze the atmosphere of the exoplanet WD 1856 b, revealing significant quantities of hydrocarbons and aerosols. These findings offer valuable information regarding the planet’s mass and thermal evolution in the context of its orbit around a white dwarf. Journal: *Nature*
- **Moderate volcanic eruptions and extreme wildfires humidify the stratosphere** — score: 0.000 Research indicates that moderate volcanic eruptions and extreme wildfires have contributed to a significant rise in stratospheric water vapor since 2005, highlighting their role as a previously unrecognized factor influencing stratospheric moisture variability through aerosol perturbations. Journal: *Nature*
- **N4-Acetylcytidine enhances synthetic mRNA translation yield and fidelity** — score: 0.000 The study investigates the impact of RNA modifications on translation efficiency and accuracy, revealing that N4-acetylcytidine significantly enhances the translation yield and fidelity of synthetic mRNAs compared to the widely used N1-methylpseudouridine. This finding suggests potential improvements in mRNA design for therapeutic applications. Journal: *Nature*
- **Food web complexity underlies biodiversity effects on ecosystem functioning** — score: 0.000 The study examines how food web complexity influences the effects of biodiversity on ecosystem functioning across various environments, including marine and freshwater ecosystems as well as soil. It reveals that intricate interactions within food webs are crucial for understanding the relationship between species diversity and the overall performance of ecosystems. Journal: *Nature*
- **Competing programs shape cortical sensorimotor–association axis development** — score: 0.000 The study presents evidence from multiple species that supports a model for cortical development where the formation of sensorimotor-to-association pathways is influenced by competing processes of induction and exclusion. These processes are driven by two distinct programs, each characterized by specific gene expression patterns, shaping the cortical sensorimotor-association axis. Journal: *Nature*
- **Correcting congenital myasthenia-associated acetylcholine receptor defects** — score: 0.000 The study employs cryogenic electron microscopy, chemical biology, and electrophysiology to analyze congenital myasthenic syndrome-associated mutant acetylcholine receptors, revealing structural and functional implications. This research aims to enhance understanding of disease mechanisms and inform the development of targeted precision therapies for affected patients. Journal: *Nature*
- **Urokodia sheds light on the origin of chelicerae and book gills of Chelicerata** — score: 0.000 The study on *Urokodia aequalis*, using microtomography and phylogenetic methods, offers new insights into the evolutionary development of key anatomical features such as chelicerae and book gills in chelicerates, based on findings from the Chengjiang biota in China. Journal: *Nature*